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ORIGINAL RESEARCH

A Deep Knowledge Distillation HeartCare Framework for Detection of Multi-label Myocardial Infarction from Multi-lead ECG Signals

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Abstract

Myocardial Infarction (MI) is the most frequent form of Coronary Heart Disease (CHD), which is the world's leading cause of illness and mortality. More than 15% of deaths per year are caused by MI. An extensively utilized diagnostic technique for cardiovascular problems is the 12-lead Electrocardiogram (ECG). Deep learning algorithms have outperformed traditional feature engineering-based techniques in terms of performance. By analyzing ECG data, deep learning models can generate clinical assessments that are comparable to those of cardiologists. The article proposes a framework for detecting and localizing MI using a convolutional neural network. The method employs a knowledge distillation approach and accepts a 12-lead ECG as input. The framework has been developed and validated using the PTB-XL dataset. The framework achieved the ability to identify and locate five super-diagnostics classes included in the PTB-XL dataset. The framework has been validated successfully, with an average accuracy of 94.25% and a micro-averaged receiver operating characteristic (ROC) of 0.95. In addition, the model is validated using a combined dataset, prepared from the Shaoxing People's Hospital (SPH) dataset, the China Physiological Signal Challenge (CPSC) 2018 dataset, and a private dataset containing ECG data from Indian patients. This allows the scientific community to assess the applicability of the proposed method to various patient groups and evaluate its performance in detecting MI. During the cross-validation process, the proposed structure achieves a mean accuracy of 95.80% and a sensitivity of 84.98%.

Keywords Myocardial infarction · Cardiovascular disease · Electrocardiogram · Knowledge distillation · Convolutional neural network

Introduction

MI is a highly severe form of Cardiovascular Diseases (CVDs). In 2016, CVDs caused the death of almost 17.9 million individuals, which represents 31% of all global deaths, as reported by the World Health Organization (WHO) in 2017 [1]. Out of these deaths, approximately 50% are caused by MI. Coronary artery disease (CAD) is a disorder called MI that occurs when one of the coronary arteries or one of its smaller branches becomes clogged. The obstruction is a result of cholesterol plaques that accumulate and constrict the coronary arteries, leading to reduced blood flow to the heart muscles. MI occurs when the heart muscles are deprived of oxygen and other necessary nutrients, leading to their eventual death. It ranks among the primary factors contributing to mortality on a global scale. A MI is diagnosed by looking at the results of an ECG and finding high levels of cardiac biomarkers in a blood sample, such as creatine kinase-MB and cardiac troponin T. Accurate diagnosis is heavily influenced by the patient's medical history and any existing comorbidities. One of the difficulties encountered by clinicians is managing the substantial volume of patient data originating from many modalities. Preventing information overload is essential for ensuring accurate diagnosis and treatment of patients. Many efforts have been made

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to develop and implement interpretable predictive models and computer-aided diagnostic solutions that can forecast important clinical outcomes in order to improve clinical therapy and provide sophisticated analytical support. The latest advancements in machine learning and deep learning models have sparked significant interest among researchers in automating the diagnosis of MI. The majority of the initial studies rely on ECG data. ECG is commonly used to confirm the diagnosis of MI, especially when certain signs of MI are present, such as T-wave inversion and ST-segment deviation. Although these findings exhibit a high level of sensitivity, they lack specificity in diagnosing MI due to the presence of identical presentations in other heart disorders such as cardiomyopathy and left bundle branch block. This could potentially result in inaccurate categorization. In order to avoid incorrect categorizations and create an accurate depiction of patients, it is necessary to have access to many types of patient data. The research community has recently shown substantial interest in utilizing historical longitudinal health information for the early diagnosis of diseases. When diagnosing MI in a clinical setting, three key factors are considered: clinical symptoms, ECG alterations, and myocardial enzyme changes. Creatine kinase-MB (CK-MB) is the earliest detectable marker of myocardial damage in the serum of individuals with MI. It exhibits a high detection rate and excellent specificity. However, the elevated levels of CK-MB in blood serum cannot be detected until 6 to 9 h after the start of MI [2]. Patients suffering from MI can be diagnosed using many approaches, including ECG [3–5], echocardiography [6], magnetic resonance imaging (MRI) [7], and by analyzing changes in cardiac biomarkers such as CK-MB [8], troponins [9], and myoglobin [10].

The American College of Cardiology and the American Heart Association have determined that "90 min" is the ideal amount of time to treat MI with interventional therapy. This specific time period has been identified as the gold standard, as it has been shown to cut the mortality risk by 40% following treatment. To avoid the delay between the diagnosis of MI and the monitoring of biomarkers, it is possible to detect a MI attack in real time by analyzing the ECG directly [11]. The 12-lead ECG readings of patients show specific abnormalities in waveform in the relevant leads, such as pathological Q-waves, ST-segment elevation, T-wave inversion, etc., depending on the location of cardiac tissue necrosis [12]. The distinctive alterations observed in a 12-lead ECG enable us to identify and differentiate various forms of myocardial infarction. Nevertheless, the variations in waveform of 12-lead ECG signals are complex due to differences in patients as well as the location and extent of MI [13]. Therefore, it continues to be a significant difficulty for cardiologists to visually recognize and explain all the different forms of 12-lead ECG signals. The essential element of early MI diagnosis is the automated recognition of pathological changes in heartbeats and the correlation between ECG leads associated with them, which facilitates precise and efficient localization.

Deep learning techniques surpassed classic feature engineering-based techniques in terms of performance. This can be attributed to the ability of deep learning models to autonomously generate distinctive feature representations from raw ECG data. These representations have the ability to capture intricate patterns and non-linear relationships that are challenging to articulate using characteristics that are manually constructed. The proposed framework presents a method for implementing multi-level knowledge distillation approaches in the field of ECG signal processing. The framework is designed to classify the multi-channel information present in a 12-lead ECG recording. Additionally, the hierarchical organization of the ECG data was utilized to acquire knowledge about patterns at the beat, rhythm, and channel levels. The goal is to increase the attention of the student network towards input locations that are susceptible to morphological changes caused by MI. In order to improve the student network's capacity to concentrate on pertinent regions for improved MI detection, this paper presents a unique way to convey this region-specific attention information from the instructor network to the student network. The proposed model primarily aims to classify and localize multi-labelled data belonging to five super diagnostic classes: 'Normal ECG (NORM)', 'Conduction Disturbance (CD)', 'ST/T Changes (STTC)', 'Myocardial Infarction (MI)', and 'Hypertrophy (HYP)'. These ECG samples are available in the PTB-XL benchmark dataset [14]. The interpretability of our model is assessed by comparing the attention visualizations with the cardiologist's criteria for identifying the MI. Additionally, in order to evaluate the generalizability of the suggested approach across various patient cohorts, the model is being cross-validated using two publicly available datasets [15, 16] in addition to a private Indian patient dataset. The primary contribution of the article is as follows:

1. We propose an innovative knowledge distillation framework designed to develop a high-precision, multi-level MI classification model using 12-lead ECG signals. This approach strategically incorporates attention mechanisms, specifically tuned to regions of the ECG that exhibit morphological changes linked to cardiac events. By integrating advanced deep learning techniques, our model significantly enhances the accuracy and granularity of MI classification, marking a substantial contribution to the field of precision cardiology and the early detection of cardiac conditions.
2. The article introduces a highly optimized student model with few parameters, specifically tailored for environments with limited computational resources. It demon-

strates that by using Multi Level Knowledge Distillation (MLKD), the student model can get important morphology-specific attention from a more advanced instructor network. The student model keeps a high degree of detection accuracy even with its drastically smaller size of 7.70 MB. It does this by efficiently using condensed knowledge to identify and examine ECG signal regions affected by myocardial infarction, thereby attaining computational efficiency without compromising performance.

3. The research thoroughly validates the capacity of the proposed framework to be applied to various populations of patients by utilizing the well-known PTB-XL dataset, as well as the SPC, CPSC 2018, and an Indian dataset. The addition of the Indian dataset is especially significant because it specifically deals with the unique ECG morphological features observed in Indian patients in contrast to those from other countries. Every ECG sample was carefully adjusted to have a consistent frequency of 100 Hz in order to maintain uniformity across all datasets. This highlights the model's capacity to be used effectively in actual clinical environments, especially in areas with diverse patient demographics.

This article is divided into various sections. Section "Literature Survey" provides a thorough analysis of the existing literature on MI classification using ECG data. In Sect. "Methods", we explain the process of pre-processing ECG signals and present the proposed approaches based on knowledge distillation. Section Results and Discussion presents and analyzes the empirical findings. Section "Conclusion" eventually provides a research conclusion.

Literature Survey

The detection and localization of clinical MI often depend on the ECG as a crucial tool because of its noninvasive nature and convenient accessibility. In recent years, numerous methods have been proposed for automated diagnosis of MI from ECG signals. These methods can broadly be classified into two categories: conventional machine learning based methods and deep learning methods. Initially, computerized algorithms for diagnosing MI focused on the task of distinguishing between MI and Healthy Control (HC) groups. The machine-learning-based methods often involve domain experts in extracting handcrafted features from ECG waveforms through suitable signal processing techniques. However, the effectiveness of these methods is restricted by the discriminative ability of the chosen features. To improve the effectiveness of their classifiers, researchers have used a variety of features, such as Wavelet Transform (WT) based features [17, 18], time-domain features [19], Principal Component Analysis (PCA) [20], Linear Discriminant Analysis (LDA) [21], Empirical Mode Decomposition (EMD) [22], statistical features [23], time and frequency domain features [24–26], Hilbert Transform (HT), and Power Spectral Density (PSD) [27]. For instance, Sun et al. [28] presented a technique for autonomously detecting MI using ECGs called latent topic Multi Instance Learning (MIL). This technique created a subject space by connecting ECGs to unlabeled heartbeats in the training dataset. For the final categorization, the topic vectors representing the ECG levels were loaded into a Support Vector Machine (SVM). In order to identify MI, Pereira et al. [29] created a technique that combines eigenspace analysis and wavelet decomposition. The ECG waveforms of varying frequencies were separated using wavelet decomposition, and then the covariance matrices were subjected to eigenspace analysis in order to calculate the eigenvalues. Patients were more accurately classified as MI cases or as HC because of these training elements. A method for detecting MI was proposed by Han et al. [30], which involved combining energy entropy with morphological features taken from 12-lead ECGs. Energy entropy was computed using derived coefficients after a thorough examination of the ECGs. Global features were then created by combining these values with regional morphological traits for precise classification. In their groundbreaking work, Sahu et al. [31] utilized Regularized Neighbourhood Component Analysis (RNCA) in conjunction with Variational Mode Decomposition (VMD) to uncover and identify mutual information. The intrinsic mode function of the VMD was used to extract statistical and non-linear features, which were then sorted using RNCA and put into an AdaBoost classifier and K-nearest Neighbors (KNN) for classification. However, these conventional techniques are less practical and reliable for therapeutic applications since they mainly rely on expert medical knowledge and necessitate a laborious feature extraction and selection process.

Deep learning approaches have gained considerable attention in recent years, particularly Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) in ECG analysis, as they involve learning complex input-to-output mappings, eliminating the necessity for handcrafted features [4, 5, 32, 33]. Hybrid techniques that combine CNN and RNN architectures to efficiently capture spatial and temporal correlations in ECG signals have been studied [34–36]. The MFB-CRNN architecture, for instance, was proposed by Liu et al. [3], It combines CNN with Long Short-Term Memory (BLSTM) to use a 12-lead ECG for MI detection. Each ECG lead's unique features are extracted by the CNN, and data from each lead's feature representation is combined by the BLSTM. Yang et al. [37] proposed a multi-scale deep learning model that incorporates a residual network combined with an attention mechanism. This approach leverages multiple layers of deep learning to extract and refine

features from ECG data, making it particularly effective in capturing the complex patterns associated with MI. Similarly, Uchiyama et al. [38] developed an end-to-end CNN model designed to process 12-lead ECG images without the need for preprocessing. Furthermore, Liu et al. [39] developed a technique for multi-lead convolutional neural network (ML-CNN) and four-lead ECG based MI detection. Lead asymmetric pooling (LAP) layers and sub-two-dimensional (2-D) convolutional layers are used to create a variety of lead multi-scale feature representations. In order to identify MI in ECG signals, Acharya et al. [40] developed an automated method based on CNN. This method achieved an accuracy of 93.53% on noisy data and 95.22% on filtered ECG data. Hao et al. [41] achieved 94.73% accuracy in MI detection from 12-lead ECG pictures using a two-dimensional CNN. In response to the growing demand for portable healthcare solutions, lightweight methods have been developed to facilitate rapid MI screening. Anwar et al. [42] introduced a method that combines a simple autoencoder with a k-nearest neighbors (k-NN) classifier. The integration of multiple machine learning techniques within hybrid models has also been explored to improve the detection and localization of MI. Ray et al. [43] proposed a hybrid deep learning model that integrates CNNs with long short-term memory networks, optimized using particle swarm optimization (PSO). This hybrid approach allows the model to capture both spatial and temporal features from ECG data, enhancing its ability to detect and localize MI effectively. ECG images were also utilized by researchers to detect and localize MI, including [44], Fang et al. [45], and Zhang et al. [46]. Novel techniques for identifying MI using ECG images include the multibranch sharing network by He et al. [47] and Ragab et al. [48]. Few studies involve MI patients, even though the majority use the PTB dataset. Research works such as [49] and [45] assess MI detection and localization on the PTB-XL dataset, which includes more MI patients, while [3, 37, 50–53] concentrate explicitly on MI detection using the PTB dataset on MI detection. It is imperative that the ECG recordings used in the training and testing sets originate from distinct individuals during the inter-patient paradigm. The intra-patient paradigm, on the other hand, permits the ECG recordings of the same patients in both sets. In inter-patient paradigms such as [49, 51, 54], inferior outcomes might arise from MI detection and localization performance misguided. Lack of interpretability and an unclear diagnostic basis for MI detection and localization are two frequent limitations of many deep learning-based investigations. Furthermore, a limited number of authors address the inter-patient scheme, which is highly relevant to therapeutic practice as it takes into consideration the unique characteristics and needs of individual patients. The majority of articles have simply examined the performance of the proposed model within individual patients using a typical ECG dataset. However, this approach does not allow for the verification of the model's usefulness and often leads to overly hopeful results.

The proposed method presents an approach to applying multi-level knowledge distillation techniques in the domain of ECG signal processing. The objective is to improve the process of categorizing students in the network by using prior knowledge in the form of attention patterns associated with regions that are responsive to morphological changes. The suggested strategy seeks to improve the student network's comprehension of specific morphological changes observed in the ECG during the onset of MI. The teacher network equips the student network with prior knowledge in the form of attention patterns, enabling this achievement. There are a lot of different ways that previous information can be added to deep learning networks. Some of these are feature engineering, putting limits on loss functions, looking at causal connections, regularization, architectural constraints, and transfer learning. The proposed approach is distinctive as it utilizes multi-level knowledge distillation to transfer morphology-specific attention between networks that carry out related but separate tasks on the same ECG data. The goal is to increase the student network's focus on input regions that are susceptible to morphological alterations caused by MI. The proposed technique introduces an approach that uses knowledge distillation to transfer region-specific attention information from a teacher network to a student network. This improves the student network's ability to focus on key areas, enhancing MI detection and localization.

Methods

Data Collection and Pre-processing

This study employs the recently released PTB-XL database, which currently stands as the most extensive publicly available dataset for electrocardiography [14]. A total of 21,837 12-lead, 10-second ECG recordings from 18,885 patients are included in the collection. 5,486 recordings in this dataset are classified as myocardial infarction (MI), whereas 9,528 recordings are classed as healthy controls (HC). The recordings are precisely 1 microvolt per least significant bit (1μV/LSB) saved in WaveForm DataBase (WFDB) format. Two sampling frequencies are provided for the data: 500 Hz and 100 Hz. The 100 Hz files are down-sampled counterparts of the 500 Hz files. Ten non-overlapping patient groups are created from the PTB-XL database in order to train, validate, and test algorithms. By using stratified sampling, all of the patient's recordings are kept within a single set. Using sets 1–8 for training, sets 9 for validation, and sets 10 for testing is the suggested method for developing a model. This

study also includes a private dataset of ECG recordings from Indian patients, the China Physiological Signal Challenge 2018 (CPSC 2018) [15], and external validation datasets from Shaoxing People's Hospital and Ningbo First Hospital (SPH) [16]. We assess our method's generalizability across various patient populations using these external datasets. Every ECG sample was divided into 1000 data points per sample and standardized to a frequency of 100 Hz for uniformity. The private dataset was gathered in cooperation with several hospitals and academic institutions in Northeastern India, and cardiologists from Apollo Hospitals in Guwahati, Assam, contributed annotations. The dataset is currently being annotated with five superclasses, namely Normal ECG (NORM), Conduction Disturbance (CD), Myocardial Infarction (MI), Hypertrophy (HYP), and ST/T change (STTC). The dataset comprises 8000 clinical 12-lead ECG records of 10 s length, where 65% were male and 35 % were female. The ECG statements used for annotation conform to the SCP-ECG standard [55]. A detailed data description of the private ECG dataset is shown in Table 1.

Architecture

The aim of this study is to tackle a challenge that arises when applying machine learning algorithms to devices that have restricted computational capabilities. Furthermore, the objective is to develop a deep learning model that is very efficient in terms of resource utilization and demands minimal computational power once deployed on the edge. This study utilizes a Knowledge distillation framework to enhance the CNN model for the purpose of generating a complete multi-level classification model. The objective is to classify 5 super diagnostics classes available in the PTB-XL dataset. The proposed framework is visually represented in Fig. 1. During the early stage, the instructor network uses labeled information to gain expertise about ECG features. In the second step, the teacher network stops acquiring new knowledge. The probability distribution q_t is generated as the result for each input X. Similarly, the student network produces the probability distribution q_s. The outputs of q_t and q_s are compared in order to calculate the distillation loss, commonly referred to as Knowledge Distillation loss.

Table 1 Summary of the Indian ECG dataset in terms of diagnostic superclasses

Records	Superclass	Description
2741	NORM	Normal ECG
1575	MI	Myocardial Infarction
1411	CD	Conduction Disturbance
764	HYP	Hypertrophy
1508	STTC	ST/T Change

Conversely, the cross-entropy loss, also known as CE loss, is calculated by comparing the outputs of q_s and their corresponding labels. Ultimately, these two losses are used to adjust the parameters of the student network.

The concept of transferring information from a big model to train a smaller model without causing a substantial impact on performance was initially introduced by Bucilua et al. (2006) [56]. Subsequently, this technique gained popularity under the name of knowledge distillation after Hinton et al. (2015) [57] introduced it using a unique method known as distillation. Knowledge distillation is a process that involves the use of a larger model, known as the teacher, and a smaller model, known as the student. The purpose of this process is to supervise and guide the student model in order to reduce the computational complexity of the final model. The complexity of deep neural networks is influenced by the number of layers and their size, which can make training and computation more challenging. Knowledge distillation is a technique that addresses this issue by compressing the knowledge from a larger ensemble model into a smaller model. This is achieved by using the class probabilities as soft targets during the training process. Neural networks generate these class probabilities using the softmax function, as represented in Eq. 1.

$$p\left(z_i, T\right) = \frac{\exp\left(z_i/T\right)}{\sum_j \exp\left(z_j/T\right)} \tag{1}$$

where z_i is logit for i^{th} class and T is temperature for controlling the importance of each soft target (default = 10); higher the T value represents softer probability distribution over classes. The distillation loss for soft logits are expressed as depicted in Eq. 2.

$$L_{\mathrm{Res}D}\left(p\left(z_t, T\right), p\left(z_s, T\right)\right) = \mathcal{L}_R\left(p\left(z_t, T\right), p\left(z_s, T\right)\right) \tag{2}$$

The expression $\mathcal{L}_R()$ represents the divergence loss of logits. The variable z_t represents the logits of the teacher, while z_s represents the logits of the student. The term $\boldsymbol{L}_R\left(p\left(z_t, T\right), p\left(z_s, T\right)\right)$ denotes the Kullback–Leibler (KL) divergence loss between the probability distributions of z_t and z_s, both conditioned on temperature T. The student network is a less complex form of the teacher network, consisting of a small number of layers and channels. The student loss is denoted by the cross-entropy loss function $\mathcal{L}_{CE}\left(y, p\left(z_s, T=1\right)\right)$, which measures the discrepancy between the true label and the soft logits produced by the student model.

At first, a teacher CNN model is used to identify and pinpoint MI. The structure of the instructor model is illustrated in Fig. 2. The model was trained using the binary cross-entropy loss function, the Adam optimizer, and a learning

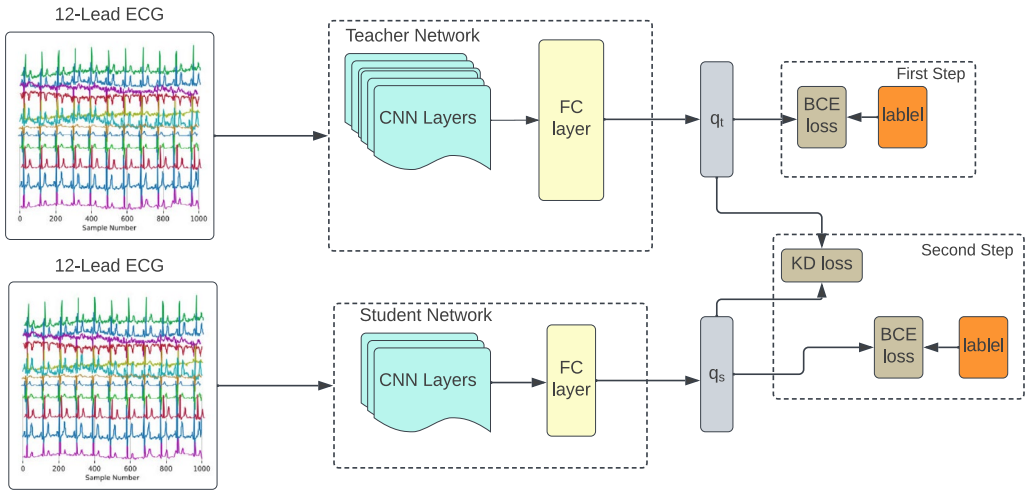


Fig. 1 Graphical overview of the proposed framework

rate of 0.001. The sigmoid activation function was used after the completely linked layer, which is why the binary cross-entropy function was used. If there is no improvement in the training process for five consecutive epochs, the training will be halted. The training was limited to a maximum of 60 epochs, using a mini batch size of 32 and employing 5-fold cross-validation. Furthermore, the model that obtained the lowest error rate on the testing dataset was selected. The total performance of MI detection and location is significantly influenced by hyperparameters such as learning rate, mini batch size, and the number of epochs. These hyperparameters are chosen and adjusted by a process of trial and error. The Table 2 provides a comprehensive overview of the proposed teacher CNN model.

Upon obtaining satisfactory results from the teacher model on both the testing and validation datasets, an endeavor was undertaken to construct a student network with the purpose of distilling the knowledge acquired from the teacher network. The student network's proposed structure is illustrated in Fig. 3. In addition to the inter-class relation, which involves transferring the relation of many classes in each instance, the prediction scores of multiple examples in each class are also instructive and valuable. These scores represent the degree of similarity between several examples and a single class. The

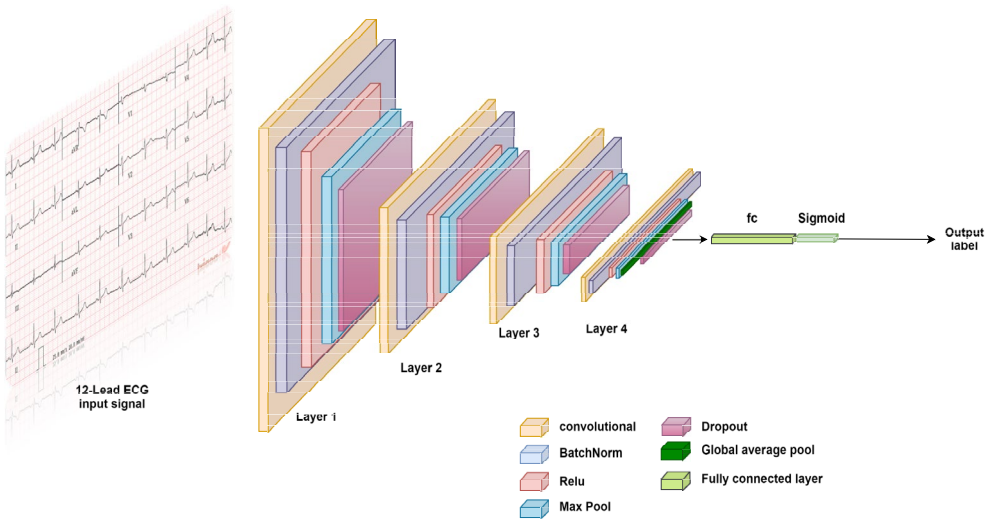


Fig. 2 Architecture of the Teacher Model

Table 2 Detailed description of the Teacher Model Architecture

Layer	Description	Parameter Count
Conv1d-1	In channels: 12, Out channels: 100, Kernel size: 10	12,100
BatchNorm1d-2	Num features: 100	200
ReLU-3	Activation function	0
MaxPool1d-4	Kernel size: 2	0
Dropout-5	Probability: 0.5	0
Conv1d-6	In channels: 100, Out channels: 250, Kernel size: 10	250,250
BatchNorm1d-7	Num features: 250	500
ReLU-8	Activation function	0
MaxPool1d-9	Kernel size: 2	0
Dropout-10	Probability: 0.5	0
Conv1d-11	In channels: 250, Out channels: 500, Kernel size: 10	1,250,500
BatchNorm1d-12	Num features: 500	1,000
ReLU-13	Activation function	0
MaxPool1d-14	Kernel size: 2	0
Dropout-15	Probability: 0.5	0
Conv1d-16	In channels: 500, Out channels: 1000, Kernel size: 10	5,001,000
BatchNorm1d-17	Num features: 1000	2,000
ReLU-18	Activation function	0
MaxPool1d-19	Kernel size: 2	0
AdaptiveAvgPool1d-20	Output size: 1	0
Dropout-21	Probability: 0.5	0
Linear-22	In features: 1000, Out features: num_classes	5,005
Sigmoid-23	Activation function	0

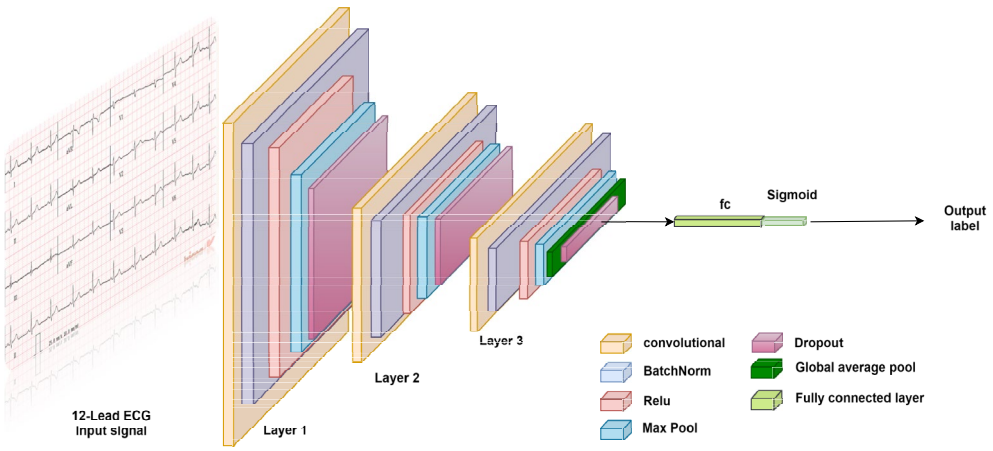


Fig. 3 Architecture of the student model

primary objective was to create a student model capable of efficiently capturing both local and global data, resulting in enhanced performance in the subsequent MI classification test. As previously said, the teacher network is trained to accurately identify and analyze the many components of ECG signals, with a particular focus on obtaining the important morphological information. Subsequently, the student network undergoes training for MI categorization and is instructed to imitate the attention patterns exhibited by the teacher network. This is accomplished by including an extra loss term that is based on

KL divergence. This loss term is used to minimize the disparity between the attention distributions of the teacher and student networks. The weights of the teacher network remain unchanged throughout the training phase of the student network. At this stage, the output of the teacher network is solely used for inference purposes. A comprehensive depiction of the student network is provided in Table 3. Figure 1 illustrate that q_t represents the output probability distribution of the trainer's network, whereas q_s represents the output probability distribution of the student network. The process of knowledge distillation is comprised of two distinct processes. Initially, the teacher network undergoes training. This network is known for its intricate structure and can also incorporate many networks that have been taught independently. The sole prerequisite is that given input X, it is capable of producing q_t, which represents the output probability distribution subsequent to the application of softmax mapping. Subsequently, the student network undergoes training, characterized by a compact parameter set and a comparatively uncomplicated network architecture. Input X can generate output q_s, which represents the resulting output probability distribution after applying softmax mapping. In the second step, the induction training is shifted from training the instructor network to inducing the student network to train. The loss function of the student network can be characterized as depicted in Eq. 3.

$$L_s(\theta_s) = (1-\alpha)\cdot \mathrm{CE}(q_s, y) + \alpha\tau^2 \cdot \mathrm{KL}\left(\mathrm{softmax}\left(\frac{q_s}{\tau}\right), \mathrm{softmax}\left(\frac{q_t}{\tau}\right)\right) \quad (3)$$

Where, $L_s(\theta_s)$ is the total knowledge distillation loss for the student model with parameters θ_s. $\mathrm{CE}(q_s, y)$ is the cross-entropy loss between the student model output q_s and the true labels y.$\mathrm{KL}\left(\mathrm{softmax}\left(\frac{q_s}{\tau}\right), \mathrm{softmax}\left(\frac{q_t}{\tau}\right)\right)$ is the Kullback-Leibler divergence between the softened outputs of the student and teacher models. α is the weighting parameter balancing the two loss components. τ is the temperature parameter. The soft target loss is scaled by $\alpha\tau^2$, illustrating the influence of the instructor model's softened outputs on the student's training process. This equation directly demonstrates the combination of the cross-entropy loss with the knowledge distillation loss.

Results and Discussion

As previously indicated, the PTB-XL dataset [14] covers a sizable cohort of disorders, which is used to assess the model. Up to two ECG specialists have annotated the raw data, and each ECG statement satisfies the SCP-ECG requirements. Additionally, each ECG statement contains several ECG statements, or labels. Only the diagnostic statements are taken into account for the initial assessment of the proposed model; the dataset contains 21430 ECG recordings total that have at least one diagnostic statement. It is possible to annotate an ECG recording with more than one diagnostic note. Hierarchical categories are given for the diagnostic statement labels in terms of 24 subclasses and 5 superclasses. The five superclass labels that indicate which

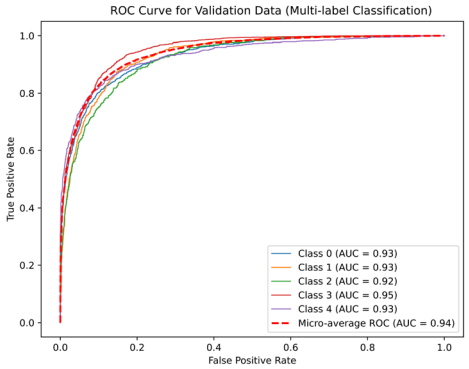
Table 3 Detailed description of the student model architecture

Layer	Description	Parameter
Conv1d-1	In channels: 12, Out channels: 50, Kernel size: 10	6,050
BatchNorm1d-2	Num features: 50	100
ReLU-3	Activation function	0
MaxPool1d-4	Kernel size: 2	0
Dropout-5	Probability: 0.3	0
Conv1d-6	In channels: 50, Out channels: 150, Kernel size: 10	75,150
BatchNorm1d-7	Num features: 150	300
ReLU-8	Activation function	0
MaxPool1d-9	Kernel size: 2	0
Dropout-10	Probability: 0.3	0
Conv1d-11	In channels: 150, Out channels: 300, Kernel size: 10	450,300
BatchNorm1d-12	Num features: 300	600
ReLU-13	Activation function	0
MaxPool1d-14	Kernel size: 2	0
Dropout-15	Probability: 0.3	0
AdaptiveAvgPool1d	Output size: 1	0
Dropout-17	Probability: 0.3	0
Linear-18	In features: 300, Out features: num_classes	301
Sigmoid-19	Activation function	0

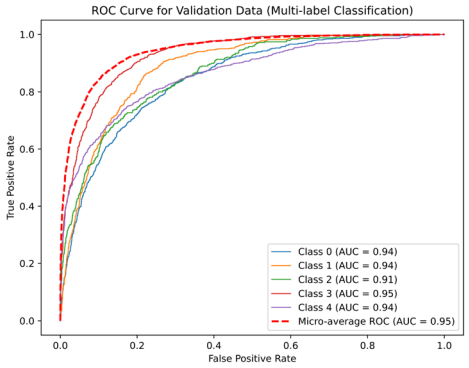
ECG recordings inside a given superclass are part of the same pathology family have been employed in our investigations. 'NORM', 'CD', 'MI', 'HYP', and 'STTC' are the superclass illness groupings. The dataset's stratified folds offer an organized way to split it into training and testing sections. Ten folds are assigned to each ECG recording: the model is trained on folds 1 through 8, the validation set is utilized on fold 9, and the test set is on fold 10. Both the training and testing sets are standardized prior to training. Metrics including the mean accuracy, maximum F1 score, and macro-averaged area under the Receiver Operating Characteristics curve (ROC-AUC) are used to assess the model's performance. Class-wise ROC-AUC scores and class-wise accuracy are also reported in addition to these measures. Finally, to validate the model's performance, special validation datasets were prepared using the SPH ECG dataset [16], the CPSC 2018 dataset [15], and one private dataset prepared using Indian-origin cardiac patients' data in collaboration with various hospitals around the north-east served as external validation datasets to assess the generalizability of our method across different patient cohorts. The class-wise ROC curve obtained for the proposed teacher and student architecture is depicted in Fig. 4a and b.

The performance metrics for the selected sub-types of MI are displayed in Tables 4 and 5. The suggested approach has outperformed previous multi-level MI classification frameworks. The proposed knowledge distillation system facilitates the transfer of morphology-specific attention maps from a teacher network, which is trained for the ECG delineation job, to a student network intended for the MI classification task. This method utilizes the common ECG signal to extract important attention patterns from the teacher network, thereby improving the performance of the student network on its assigned task. As a result, the student network's performance metrics have improved compared to the teacher model.

As mentioned earlier, to evaluate the model's performance in MI detection, this study also utilizes a special validation dataset to assess the generalizability of the proposed method across different patient cohorts. With these three datasets, the proposed method demonstrated high performance, indicating strong generalization capabilities. While SPH Dataset [16] and CPSC 2018 [15] datasets are widely used in research, we recognize the importance of validating our findings on practical data. To address this aspect, our study also incorporates Indian patient datasets for the validation of the proposed framework. The ROC-AUC curve for the proposed framework obtained while validating with an external dataset is depicted in Fig. 5. The proposed framework achieved a mean accuracy of 95.80% and a sensitivity of 84.98% on the validation datasets. The instructor model, as proposed, has a total size of approximately 38.13*MB*, whereas the student model has a total size of approximately



(a) ROC-AUC curve for Teacher CNN model



(b) ROC-AUC curve for Student CNN model

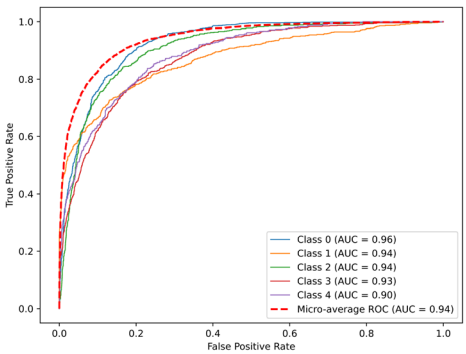
Fig. 4 ROC-AUC Curves obtained for both model using validation data from PTB-XL dataset [14]

Table 4 Class-wise ROC-AUC Comparison with existing framework

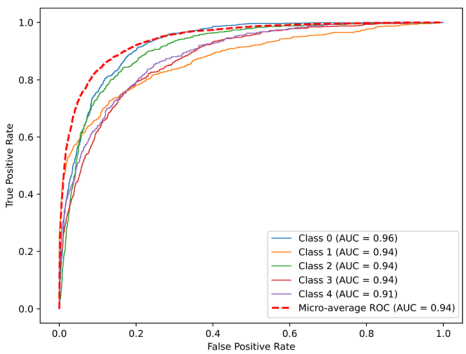
	NORM *(Class 0)*	MI *(Class 1)*	STTC *(Class 2)*	HYP *(Class 3)*	CD *(Class 4)*
Mousavi et al. [58]	0.9096	0.8708	0.8817	0.8093	0.8555
ECGNet [59]	0.9336	0.9021	0.9205	0.8842	0.9102
Rajpurkar et al. [60]	0.9309	0.9066	0.9306	0.9185	0.8910
IMLE-Net [61]	0.9450	0.9244	0.9354	0.8834	0.9196
Geng et al. [62]	0.904	0.877	0.914	0.911	0.931
Wen et al. [63]	0.93	0.90	0.93	0.95	0.93
Proposed [Teacher]	0.9356	0.9374	0.9231	0.9562	0.9312
Proposed [Student]	0.9464	0.9463	0.9177	0.9572	0.9468

Table 5 Class-wise accuracy comparison with existing framework

	NORM *(Class 0)*	MI *(Class 1)*	STTC *(Class 2)*	HYP *(Class 3)*	CD *(Class 4)*
Mousavi et al. [58]	81.78	82.47	83.26	87.19	86.26
ECGNet [59]	85.89	84.60	86.87	90.66	88.71
Rajpurkar et al. [60]	85.94	85.94	88.90	88.76	90.01
IMLE-Net [61]	87.10	87.97	89.08	91.58	88.53
Geng et al. [62]	90.50	91.20	86.90	88.30	86.80
Wen et al. [63]	84.08	84.70	80.71	82.05	80.64
Proposed [Teacher]	95.00	92.50	93.75	94.25	91.75
Proposed [Student]	95.50	93.75	94.00	94.75	93.25



(a) ROC-AUC curve for Teacher CNN model



(b) ROC-AUC curve for Student CNN model

Fig. 5 ROC-AUC Curves obtained for both model using external validation dataset prepeared using SPH ECG dataset [16], the CPSC 2018 dataset [15], and one private dataset prepared using Indian-origin cardiac patients' data in collaboration with various hospitals around the North-East region

7.70*MB*. Despite the difference in size, both models exhibit the same level of performance. Networks are generally viewed as opaque structures in deep learning. In computer vision, Grad-CAM [64] is a commonly used method for visually representing image data and highlighting the particular features that convolution identifies. Grad-CAM is a method that runs the gradient of a target feature through the last convolution layer to produce a weighted activation map. The key areas of a picture that are crucial for predicting the target category are indicated on this map. This method was used in this work to visually depict the features of the ECG, which required adjustments to account for the signal's one-dimensionality. The suggested network focuses on analyzing specific wave clusters on specific ECG leads, as depicted in Fig. 6. It mainly focuses on the QRS waves on leads II, III, aVR, V1, V2, and aVF. Additionally, focus more on leads I, aVL, V3, and V4's P and T waves. To further validate our method, we thoroughly evaluated its performance against six cutting-edge deep learning architectures, as illustrated in Tables 4 and 5. In particular, Table 4 presents a comparison of the class-wise ROC-AUC with existing frameworks, while Table 5 offers a comparison of the class-wise accuracy with existing frameworks. The present cutting-edge approaches primarily depend on spatio-temporal representations of ECG data, demonstrating very competitive performance across three different datasets. However, the exclusively data-driven architecture of these systems significantly hinders their performance and lacks interpretability. In contrast, the proposed framework consistently outperforms current models by combining a unique morphological attention mechanism with attention distillation. By integrating domain-specific context into the model via attention, the proposed method successfully captures both temporal and morphological patterns in the ECG signals, resulting in improved feature representation and decision-making functions. The thorough empirical evaluation proves that the suggested framework is the most effective way to find MI using 12-lead ECG signals.

Conclusion

This study developed an interpretable multi-level knowledge distribution architecture for multi-channel ECG classification. The preliminary evaluation of the suggested framework on the PTB-XL dataset shows that it performs better than a number of the current models. This study also uses a special validation dataset, which includes the SPH dataset, CPSC 2018 datasets, and an Indian patient dataset, to validate the proposed framework, assess the generalizability of the proposed method across different patient cohorts, and evaluate the model's performance in MI detection. To comprehend how the model's performance varied with different combinations of ECG channels, an

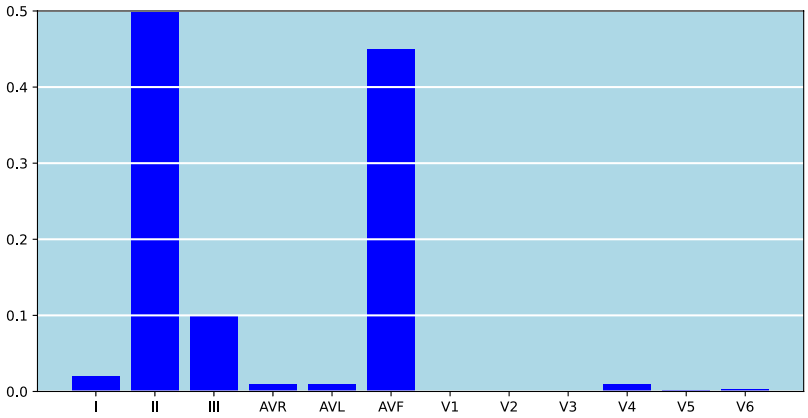


Fig. 6 Visualization of normalized attention scores to various lead of ECG signals

ablation study was conducted.The attention scores that were produced were shown and contrasted with the criteria that cardiologists use to identify two different kinds of myocardial infarction. The interpretability feature of the suggested model enables a clearer comprehension of its utility in clinical decision-making in real-world scenarios. The interpretability of the model can be assessed in the future against a variety of cardiac diseases. Additionally, find more efficient ways to handle issues with data imbalance, numerous labels, and noise in ECG dataset classification. The majority of real ECG readings are noisy because of medical hardware. Some arrhythmia cases are uncommon and will result in imbalance issues; other cases have arrhythmia issues of their own. Our research revealed that the imbalance issue in ECG datasets has a significant negative impact on our network's performance. When the multi-label issue and the imbalance issue arise simultaneously in an ECG dataset, it turns into a crossing issue that is more challenging to resolve. Future efforts will be directed at integrating the suggested framework into Field Programmable Gate Arrays (FPGAs) devices.

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Data Availibility The publicly available ECG data is collected from the Physionet Repository. Further, the validation ECG samples collected from the medical institution are not shareable or permitted to be shared to protect the privacy of the patients.

Code Availability On request, the codes used in the experiments might be provided.

Materials Availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no Conflict of interest.

Ethics approval and consent to participate Not applicable.

Consent for publication After reading the paper, each author gave their approval for it to be published.

References

1. Roth GA, Mensah GA, Johnson CO, Addolorato G, Ammirati E, Baddour LM, Barengo NC, Beaton AZ, Benjamin EJ, Benziger CP, et al. Global burden of cardiovascular diseases and risk factors, 1990–2019: update from the gbd 2019 study. J Am Coll Cardiol. 2020;76(25):2982–3021.
2. Costa CM, Silva IS, Sousa RD, Hortegal RA, Regis CDM. The association between reconstructed phase space and artificial neural networks for vectorcardiographic recognition of myocardial infarction. J Electrocardiol. 2018;51(3):443–9.
3. Liu W, Wang F, Huang Q, Chang S, Wang H, He J. Mfb-cbrnn: a hybrid network for mi detection using 12-lead ecgs. IEEE J Biomed Health Inform. 2019;24(2):503–14.
4. Hammad M, Alkinani MH, Gupta BB, Abd El-Latif AA. Myocardial infarction detection based on deep neural network on imbalanced data. Multimed Syst, 2022; pp 1–13
5. Makimoto H, Höckmann M, Lin T, Glöckner D, Gerguri S, Clasen L, Schmidt J, Assadi-Schmidt A, Bejinariu A, Müller P, et al. Performance of a convolutional neural network derived from an ecg database in recognizing myocardial infarction. Sci Rep. 2020;10(1):8445.
6. Peels CH, Visser CA, Kupper AJF, Visser FC, Roos JP. Usefulness of two-dimensional echocardiography for immediate detection of myocardial ischemia in the emergency room. Am J Cardiol. 1990;65(11):687–91.

7. Tsai D-Y, Kojima K. Measurements of texture features of medical images and its application to computer-aided diagnosis in cardiomyopathy. Measurement. 2005;37(3):284–92.
8. De Winter RJ, Koster RW, Sturk A, Sanders GT. Value of myoglobin, troponin t, and ck-mbmass in ruling out an acute myocardial infarction in the emergency room. Circulation. 1995;92(12):3401–7.
9. McCord J, Nowak RM, McCullough PA, Foreback C, Borzak S, Tokarski G, Tomlanovich MC, Jacobsen G, Weaver WD. Ninety-minute exclusion of acute myocardial infarction by use of quantitative point-of-care testing of myoglobin and troponin i. Circulation. 2001;104(13):1483–8.
10. Plebani M, Zaninotto M. Diagnostic strategies using myoglobin measurement in myocardial infarction. Clin Chim Acta. 1998;272(1):69–77.
11. Thygesen K, Alpert JS, Jaffe AS, Chaitman BR, Bax JJ, Morrow DA, White HD, Myocardial Infarction EG. Fourth universal definition of myocardial infarction (2018). Circulation. 2018;138(20):618–51.
12. Arini PD, Valverde ER. Beat-to-beat electrocardiographic analysis of ventricular repolarization variability in patients after myocardial infarction. J Electrocardiol. 2016;49(2):206–13.
13. Ansari S, Farzaneh N, Duda M, Horan K, Andersson HB, Goldberger ZD, Nallamothu BK, Najarian K. A review of automated methods for detection of myocardial ischemia and infarction using electrocardiogram and electronic health records. IEEE Rev Biomed Eng. 2017;10:264–98.
14. Wagner P, Strodthoff N, Bousseljot R-D, Kreiseler D, Lunze FI, Samek W, Schaeffter T. Ptb-xl, a large publicly available electrocardiography dataset. Sci Data. 2020;7(1):1–15.
15. Liu F, Liu C, Zhao L, Zhang X, Wu X, Xu X, Liu Y, Ma C, Wei S, He Z, et al. An open access database for evaluating the algorithms of electrocardiogram rhythm and morphology abnormality detection. J Med Imag Health Inform. 2018;8(7):1368–73.
16. Zheng J, Zhang J, Danioko S, Yao H, Guo H, Rakovski C. A 12-lead electrocardiogram database for arrhythmia research covering more than 10,000 patients. Sci Data. 2020;7(1):48.
17. Parmar R, Yadav K, Anand G, Trivedi G. An snn inspired area and power efficient vlsi architecture of myocardial infarction classifier for wearable devices. IEEE Transactions on Circuits and Systems II: Express Briefs; 2024.
18. Lian S, Gao Z, Wang H, Liu X, Xu L, Liu H, Zhang H. Frequency-enhanced geometric-constrained reconstruction for localizing myocardial infarction in 12-lead electrocardiograms. IEEE Trans Biomed Eng, 2024
19. Jiang M, Bian F, Zhang J, Huang T, Xia L, Chu Y, Wang Z, Jiang J. Myocardial infarction detection method based on the continuous t-wave area feature and multi-lead-fusion deep features. Physiol Meas, 2024
20. Borra S.R, Nayana D.R.G.A, Srinidhi S, Bhavana S, Nishitha P, Sahithi V. Cadnet: cardiac arrhythmia detection and classification using unified principal component analysis and 1d-cnn model. Res Biomed Eng, 2024; pp 1–13
21. Turnip A, Hidayat D, Hamonangan JG, Novita D, Syafei NS, Fuady MI, Zaidan MGS, Montoya TRD et al., Classification of ecg signals using decision trees and linear discriminant analysis. In: 2024 IEEE International Conference on Artificial Intelligence and Mechatronics Systems (AIMS), pp. 1–6, 2024. IEEE
22. Sun Q, Wang L, Li J, Liang C, Yang J, Chen Y, Wang C. Multiphase ecg dynamic features for detecting myocardial ischemia and identifying its etiology using deterministic learning. Biomed Signal Process Control. 2024;88: 105498.
23. Di Vito L, Di Giusto F, Bruscoli F, Scalone G, Silenzi S, Mariani L, Selimi A, Delfino D, Grossi P. Recurrent events after acute st-segment elevation myocardial infarction: predictors and features of plaque progression and stent failure. Coronary Artery Disease, 10–1097, 2024
24. Jain P, Deshmukh A, Padole H. Design of an integrated myocardial infarction detection model using ecg connectivity features and multivariate time series classification. IEEE Access (2024)
25. Salem O, Alsubhi K, Mehaoua A, Boutaba R. Markov models for anomaly detection in wireless body area networks for secure health monitoring. IEEE J Sel Areas Commun. 2020;39(2):526–40.
26. Liao Y, Xiang Y, Zhao Z, Ai D. Bayesian mixed-effect higher-order hidden markov models with applications to predictive healthcare using electronic health records. IISE Trans, 2024; 1–13
27. Vinay N, Vidyasagar K, Rohith S, Dayananda P, Supreeth S, Bharathi S. An rnn-bi lstm based multi decision gan approach for the recognition of cardiovascular disease (cvd) from heart beat sound: a feature optimization process. IEEE Access, 2024
28. Sun L, Lu Y, Yang K, Li S. Ecg analysis using multiple instance learning for myocardial infarction detection. IEEE Trans Biomed Eng. 2012;59(12):3348–56. https://doi.org/10.1109/TBME.2012.2213597. (**Cited by: 200.**).
29. Pereira H, Daimiwal N. Analysis of features for myocardial infarction and healthy patients based on wavelet. In: 2016 Conference on Advances in Signal Processing (CASP), pp. 164–169, 2016) https://doi.org/10.1109/CASP.2016.7746158
30. Han C, Shi L. Automated interpretable detection of myocardial infarction fusing energy entropy and morphological features. Comput Methods Progr Biomed. 2019;175:9–23. https://doi.org/10.1016/j.cmpb.2019.03.012.
31. Sahu G, Ray KC. An efficient method for detection and localization of myocardial infarction. IEEE Trans Instrum Meas. 2021;71:1–12.
32. Prabhakararao E, Dandapat S. Attentive rnn-based network to fuse 12-lead ecg and clinical features for improved myocardial infarction diagnosis. IEEE Signal Process Lett. 2020;27:2029–33.
33. Amini S.S, Mandala S, Pramudyo M. Myocardial infarction prediction using rnn deep learning algorithm on phonocardiogram signals. In: 2023 11th International Conference on Information and Communication Technology (ICoICT), pp 92–97, 2023. IEEE
34. Feng K, Pi X, Liu H, Sun K. Myocardial infarction classification based on convolutional neural network and recurrent neural network. Appl Sci. 2019;9(9):1879.
35. Hasbullah S, Mohd Zahid MS, Mandala S. Detection of myocardial infarction using hybrid models of convolutional neural network and recurrent neural network. BioMedInformatics. 2023;3(2):478–92.
36. Borah B.B, Baruah S.M.B, Kakati D, Roy S. Cvd prediction on micro-controller: Ecg morphology learning approach. Innov Syst Softw Eng, pp 1–11, 2022
37. Cao Y, Liu W, Zhang S, Xu L, Zhu B, Cui H, Geng N, Han H, Greenwald SE. Detection and localization of myocardial infarction based on multi-scale resnet and attention mechanism. Front Physiol. 2022;13: 783184.
38. Uchiyama R, Okada Y, Kakizaki R, Tomioka S. End-to-end convolutional neural network model to detect and localize myocardial infarction using 12-lead ecg images without preprocessing. Bioengineering. 2022;9(9):430.
39. Liu W, Zhang M, Zhang Y, Liao Y, Huang Q, Chang S, Wang H, He J. Real-time multilead convolutional neural network for myocardial infarction detection. IEEE J Biomed Health Inform. 2017;22(5):1434–44.
40. Acharya UR, Fujita H, Oh SL, Hagiwara Y, Tan JH, Adam M. Application of deep convolutional neural network for automated detection of myocardial infarction using ecg signals. Inf Sci. 2017;415:190–8.

41. Hao P, Gao X, Li Z, Zhang J, Wu F, Bai C. Multi-branch fusion network for myocardial infarction screening from 12-lead ecg images. Comput Methods Progr Biomed. 2020;184: 105286.
42. Anwar SMS, Pal D, Mukhopadhyay S, Gupta R. A lightweight method of myocardial infarction detection and localization from single lead ecg features using machine learning approach. IEEE Sensors Lett, 2024
43. Sahu G, Ray KC. Pso optimized hybrid deep learning model for detection and localization of myocardial infarction. IEEE Sensors J, 2024
44. Jikui L, Ruxin W, Bo W, Zengding L, Fen M, Ye L. Myocardial infarction detection and localization with electrocardiogram based on convolutional neural network. Chin J Electron. 2021;30(5):833–42.
45. Fang R, Lu C-C, Chuang C-T, Chang W-H. A visually interpretable detection method combines 3-d ecg with a multi-vgg neural network for myocardial infarction identification. Comput Methods Progr Biomed. 2022;219: 106762.
46. Zhang J, Liu M, Xiong P, Du H, Zhang H, Sun G, Hou Z, Liu X. Automated localization of myocardial infarction of image-based multilead ecg tensor with tucker2 decomposition. IEEE Trans Instrum Meas. 2021;71:1–15.
47. He C, Liu M, Xiong P, Yang J, Du H, Xu J, Hou Z, Liu X. Localization of myocardial infarction using a multi-branch weight sharing network based on 2-d vectorcardiogram. Eng Appl Artif Intell. 2022;116: 105428.
48. Attallah O, Ragab DA. Auto-myin: Automatic diagnosis of myocardial infarction via multiple glcms, cnns, and svms. Biomed Signal Process Control. 2023;80: 104273.
49. Liu W, Ji J, Chang S, Wang H, He J, Huang Q. Evombn: evolving multi-branch networks on myocardial infarction diagnosis using 12-lead electrocardiograms. Biosensors. 2021;12(1):15.
50. Jian J-Z, Ger T-R, Lai H-H, Ku C-M, Chen C-A, Abu PAR, Chen S-L. Detection of myocardial infarction using ecg and multi-scale feature concatenate. Sensors. 2021;21(5):1906.
51. Han C, Shi L. Ml-resnet: a novel network to detect and locate myocardial infarction using 12 leads ecg. Comput Methods Progr Biomed. 2020;185: 105138.
52. Martin H, Morar U, Izquierdo W, Cabrerizo M, Cabrera A, Adjouadi M. Real-time frequency-independent single-lead and single-beat myocardial infarction detection. Artif Intell Med. 2021;121: 102179.
53. Rai HM, Chatterjee K. Hybrid cnn-lstm deep learning model and ensemble technique for automatic detection of myocardial infarction using big ecg data. Appl Intell. 2022;52(5):5366–84.
54. Fu L, Lu B, Nie B, Peng Z, Liu H, Pi X. Hybrid network with attention mechanism for detection and location of myocardial infarction based on 12-lead electrocardiogram signals. Sensors. 2020;20(4):1020.
55. Liu H, Chen D, Chen D, Zhang X, Li H, Bian L, Shu M, Wang Y. A large-scale multi-label 12-lead electrocardiogram database with standardized diagnostic statements. Sci Data. 2022;9(1):272.
56. Buciluǎ C, Caruana R, Niculescu-Mizil A. Model compression. In: Proceedings of the 12th ACM SIGKDD International conference on knowledge discovery and data mining, pp 535–541, 2006
57. Hinton G, Vinyals O, Dean J. Distilling the knowledge in a neural network. arXiv preprint arXiv:1503.02531 (2015)
58. Mousavi S, Afghah F. Inter- and intra- patient ecg heartbeat classification for arrhythmia detection: a sequence to sequence deep learning approach. In: ICASSP 2019 - 2019 IEEE international conference on acoustics, speech and signal processing (ICASSP), pp 1308–1312, 2019. https://doi.org/10.1109/ICASSP.2019.8683140
59. Murugesan B, Ravichandran V, Ram K, Preejith SP, Joseph J, Shankaranarayana SM, Sivaprakasam M. Ecgnet: Deep network for arrhythmia classification. In: 2018 IEEE International Symposium on Medical Measurements and Applications (MeMeA), pp 1–6, 2018. https://doi.org/10.1109/MeMeA.2018.8438739
60. Hannun AY, Rajpurkar P, Haghpanahi M, Tison GH, Bourn C, Turakhia MP, Ng AY. Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network. Nat Med. 2019;25(1):65–9.
61. Reddy L, Talwar V, Alle S, Bapi RS, Priyakumar UD. Imle-net: an interpretable multi-level multi-channel model for ecg classification. In: 2021 IEEE International Conference on Systems, Man, and Cybernetics (SMC), pp 1068–1074, 2021. IEEE
62. Geng Q, Liu H, Gao T, Liu R, Chen C, Zhu Q, Shu M. An ecg classification method based on multi-task learning and cot attention mechanism. Healthcare. 2023. https://doi.org/10.3390/healthcare11071000.
63. Wen W, Zhang H, Wang Z, Gao X, Wu P, Lin J, Zeng N. Enhanced multi-label cardiology diagnosis with channel-wise recurrent fusion. Comput Biol Med. 2024;171: 108210.
64. Selvaraju RR, Cogswell M, Das A, Vedantam R, Parikh D, Batra D. Grad-cam: visual explanations from deep networks via gradient-based localization. In: Proceedings of the IEEE International Conference on Computer Vision, pp 618–626, 2017

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